Advanced Financial Management 2025-2026

Case

Procedure

- Use the font: Times New Roman, 12pt, 1.5 line spacing. Margins of at least 2 cm (almost any standard text editor will have margins over 2 cm as a the 'normal' setting).
- Hand in 3 files: a report that answers the questions, a data-file with the data and a file that shows your code.
- Documents in pdf (report), xls/xlsx (data-file) and r (code) are accepted.
- Maximum no. of pages for the report (excluding references and cover page): 6
- If you hand in a document that is more than 6 pages, only the first 6 pages will be read.
- Each question has to be answered on a separate page. Each page should have a graph or table and text that adds explanation to the graph or table. The graph or table should take up at least 33% of the page (inside the margins).
- The assignment needs to be completed in Google Docs. This is such that we can keep track of the version history. All text needs to be written in Google Docs. This means that if text is copied from an outside source, this text will be disregarded and the section that has this text will be graded with a zero.
- After completing the assignment in Google Docs, export the document as a pdf file for uploading to Canvas. The final version in Google Docs and the pdf file need to match exactly.
- The text in the pdf document needs to be selectable as text. DO NOT import the text as a picture. Exporting to pdf from Google Docs will ensure that the text in the pdf document is selectable as text.
- Graphs in the document do not need to be selectable as text. Graphs can be scanned or imported from other files.
- Do not plagiarize. Working together on the case is fine and encouraged, copying each other's answers (or answers from any other source) is not. If plagiarism is detected or suspected, this will be communicated to the Examination Board. Make sure your work is your own.

- The use of AI is not allowed in this course, also not for editing grammar. The whole point of the case is for you to think about the materials we are studying in the course. Outsourcing your thinking and creativity to an AI is fine after your studies, but not during your studies. After all, how are you going to learn how to think better than an AI if you never practice? If generative AI in a section is suspected beyond a reasonable doubt by both instructors, this section will be graded with a zero. AI suspicion = usually means: Vague, non-specific and meandering text which sounds good but does not directly address the question. Be precise, be specific and do your own work.

General instructions:

- Explain what you are doing. Don't just jot down a couple of numbers and expect us to be able to tell what's going on. Do not simply dump Excel- or R-output in your file and expect us to understand. Explain your reasoning, write out formulas before you fill in the numbers. Make sure graphs and tables are easily understood without having to refer to text outside of the graph or table. In short: Show your thought process, don't assume we can look in your head.
- Make sure all the essential information is in the main body of the report. Adding appendices with additional information is fine. However: Assume appendices will not be read, they are treated as if they do not exist. The need-to-know has to be in the text, nice-to-know can go in the appendix.
- A cover page with your names, student numbers and link to the Google Doc is fine. However, a table of contents is not necessary for a 6-page report.
- The references section does not count towards the page limit. Use as many references as you want.

Data and code:

- Treat the data-file and code as an appendix. As such: Make sure the need-to-know is in the report, assume the Excel-file and R-file will not be read.
- That said: Layout of the data-file and code still matters. Make sure the data-file and code are easily understood and readable to an outsider.

Preparation

1. Add the link to the Google Doc (link such that anyone with the link can edit the document) at the top of the document you hand in. If the Google Doc link is not provided (or the link does not have editor privileges for anyone to edit the document), the assignment will not be graded.
2. Download daily historical stock returns for 10 stocks from the S&P500 and the S&P500 itself for the last ten years – up until the end of 2024.[1] (you can use whatever data provider you like, but CRSP is probably easiest – start at https://libguides.vu.nl/finding-data/ to see how you can get access).

[1] If you're interested in a company that has been listed less than 10 years, you can use the date the stock has been listed as the start date. However, it can't be shorter than 5 years. If that's the case, choose another stock.

Page 1: Efficient frontier (has to be completed fully in Excel)

Intermediate steps (do not need to be presented in the final document)

1. For all stocks, calculate daily average returns and daily volatilities and annualize them.
2. Take two of your selected stocks and make a graph with the annualized returns on the vertical axis and the annualized volatilities on the horizontal axis.
3. Estimate an efficient frontier based on portfolios of these two stocks. Show the efficient frontier in the graph as a dark blue line.
4. Now take the stocks from the previous question and add a third stock. Now you have three two-stock combinations (A+B, A+C, B+C). Show efficient frontiers based on these three two-stock combinations as a light-blue, purple and green line.
5. Take the three stocks from the previous two questions and estimate an efficient frontier based on portfolios of these three stocks. Show the efficient frontier in the same graph as a red line (this means that the graph should now show 4 efficient frontiers).
6. Look up the current risk-free rate. Draw two capital market lines: one based on one of the two-stock portfolios and one based on the three-stock portfolio. Use the colors of the efficient frontiers as the colors for the capital market lines.

Deliverable:

7. Add the graph to your report. Make sure the graph takes up at least one third of the page so that it is readable. In the text, explain the graph that you created. Why do the graphs have the shape that they do? Why is one higher/lower compared to the other? What can be learned from this exercise?

Page 2: Beta estimation

Intermediate steps (do not need to be presented in the final document)

1. Estimate beta for one of the stocks using the last 2 years of data.
2. Re-estimate rolling-window betas using 2 years of returns for each of your stocks (going back 10 years in total) and put them in line graphs. This means you will have graphs with daily frequencies that span 8 years.
3. Out of all of these stocks, pick one stock that has big changes in beta over time and present the rolling-window betas for this one stock in a line-graph. Add vertical lines to indicate important events that have affected the beta for this stock.

Deliverable:

4. Add the graph from the last step to your report. Make sure the graph takes up at least one third of the page so that it is readable. In the text, explain the graph that you created. Observe how the company's beta changes over time. Explain how and why beta has changed over time. Which events were most important for this change in beta? Add sources.

Page 3: Factor models

Intermediate steps (do not need to be presented in the final document) – Exercise can be completed in R.

1. Go to Kenneth French's website[2] and download data on daily Fama/French 5 Factors. Using this dataset, estimate the CAPM from the beginning of your sample until the end of 2023 for each of your stocks.
2. Using your estimates from the previous question and the Kenneth French dataset, estimate expected returns over the data in your sample since the beginning of the year 2024 to the latest available date (so: estimate returns for a 1-, 3- and 5-factor model). Using the share price at the beginning of the year, make a graph with the expected price over the year based on the 1-, 3- and 5-factor model, and the actual share price. Choose one stock for which you have interesting findings to present the graph of.

Deliverable:

3. Add the graph to your report. Make sure the graph takes up at least one third of the page and that it is readable. In the text, explain the graph that you created. Compare the expected returns for each of the three models with the actual returns by looking at the overall (expected) cumulative returns over 2024. Which of the models performs best? How can this be explained?

[2] https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html

Page 4: Project Finance

Take the company for which you estimated the median beta over the period Jan 2023- Dec 2024 for the second page of your document. Suppose this company is expanding its production in a new project. To enable the expansion, they are contemplating investing $1.7 billion in a new plant with an expected 10-year life, which is depreciated linearly over the lifetime of the project. The anticipated revenues would be $900 million in the first year of operation and grow by 10% for each of the next two years and 6% in the seven years afterwards. COGS are estimated at 60% of revenues, and SGA expenses are estimated to remain constant at $150 million throughout the project. Net working capital needs are 5% of sales. Assume a 25% tax rate. You have been asked to evaluate the new project.

Intermediate steps (do not need to be presented in the final document)

1. Create a financial projection table. Show years as columns, and relevant items as rows (you can find examples of financial projections in chapter 18 of Berk&DeMarzo). Make sure the cash flows are clear, and that it's clear how you came to these cash flows.
2. Assume the project has the same costs of capital as the existing firm. Calculate the relevant discount factors, show them in the table, and show the discounted cash flows in the table. Calculate the NPV of the project.
3. Suppose the project was fully funded with equity but otherwise has the same operational risk as the existing firm. Calculate the relevant discount factors, show them in the table, and show the discounted cash flows in the table. Calculate the NPV of the project.
4. Now suppose the D/E ratio of the project is double that of the existing firm. You may assume that this does not affect the debt costs of the project, which are still the same as for the existing firm. Calculate the relevant discount factors, show them in the table and show the discounted cash flows in the table. Calculate the NPV of the project.

Deliverable:

5. Show the financial projection table with the discount factors and discounted cash flows. Make sure the table takes up at least one third of the page so that it is readable. In the text, explain the table that you created. Why are the cash flows what they are? Why are the discount factors what they are? How did you calculate discount rates? How do the NPVs of the project compare under different capital structures?

Page 5: Short-term finance

Intermediate steps (do not need to be presented in the final document)

1. For each of the 10 companies for which you downloaded stock returns, calculate the inventory days, accounts payable days, and accounts receivable days. Note that Compustat (accessible through the VU - https://libguides.vu.nl/finding-data/) has book data for most US firms.
2. Put the inventory days, accounts payable days, and accounts receivable days in a bar graph (stack inventory days, accounts payable days, and accounts receivable days on top of each other), and add the cash conversion cycle as a line graph in that same bar graph.
3. Sort the graph by cash conversion cycle: Lowest CCC left, highest CCC right.

Deliverable:

4. Show the graph. Make sure the graph takes up at least one third of the page so that it is readable. In the text, explain the graph that you created. Specifically, explain for the companies with the lowest and highest CCC why they have such low or high CCCs. Can anything be improved for any of these companies? Make concrete suggestions (e.g., stating 'inventories should be lowered' is meaningless unless you explain how this might actually work) and use sources.

Page 6: Event study Liberation Day

Intermediate steps (do not need to be presented in the final document)

1. For each of your 10 selected stocks, estimate beta over the entire data sample you've collected.
2. Look up the risk-free rate and return on the S&P500 around Liberation Day (April 2, 2025) for 30 days before and after the event, and estimate expected daily returns for each stock based on the CAPM.
3. Calculate daily abnormal returns for each stock (actual returns minus expected returns) and calculate the cumulative abnormal returns by day for each stock.
4. Make a graph with the cumulative abnormal returns by day for each stock (so: a graph with days on the horizontal axis, and cumulative abnormal returns on the vertical axis), starting at least 30 working days before Liberation Day and 30 working days afterwards.

Deliverable:

5. Add the graph to your report. Make sure the graph takes up at least one third of the page so that it is readable. In the text, explain the graph that you created. Explain which companies were hit especially hard and which were not. Suppose you are asked to come up with a research question surrounding Liberation Day: Could you come up with a variable that measures which companies were hit especially hard?

Rubric

Criteria	Excellent (9.0-10)	Great (8.0-8.5)	Good (7.0-7.5)	Sufficient (5.5-6.5)	Insufficient (5.0 or below)
Efficient Frontier	Graph includes all required efficient frontiers and capital market lines, correctly labeled and colored; text clearly explains shapes, comparisons, and insights with strong reasoning.	Graph includes most elements with correct colors; explanation is clear but lacks some depth.	Graph and explanation are present but limited in analysis or clarity; minor omissions.	Graph is incomplete or unclear; explanation is superficial or partially incorrect.	Missing graph or explanation; major errors in concept.
Beta Estimation	Rolling beta graph correctly shows events and trends; explanation clearly connects changes to events with well-supported reasoning and references.	Graph and vertical lines included; explanation connects events to beta but lacks strong depth or sources.	Graph is correct but explanation is basic; event analysis limited.	Incomplete graph or explanation; lacks clear connection to events.	Missing or incorrect graph; no meaningful explanation.
Factor Models	All three models (1-, 3-, 5-factor) shown and compared; analysis explains performance differences thoroughly with strong interpretation.	All models included; explanation of differences is clear but somewhat limited in depth.	Models included but analysis is minimal; lacks deep interpretation.	Graph or models incomplete; explanation unclear or missing comparisons.	Missing graph or incorrect models; no analysis provided.
Project Finance	Projection table fully correct, clear and well-formatted; text explains cash flows, discount factors, NPVs under all structures with solid reasoning.	Table and explanation mostly correct; some minor issues in clarity or depth.	Table present but with errors or unclear layout; explanation basic.	Table or explanation incomplete; weak justification for results.	Missing table or largely incorrect; no explanation.
Short-term Finance	Bar and line graph fully correct and sorted; explanation gives strong analysis of CCC extremes with practical, well-sourced suggestions.	Graph correct and sorted; explanation includes some improvement ideas but limited detail.	Graph and explanation present but weak analysis or generic suggestions.	Graph or explanation incomplete; superficial or unclear.	Missing or incorrect graph; no meaningful explanation.
Event Study	Graph correctly shows cumulative abnormal returns for all stocks; explanation identifies impacted companies and suggests a meaningful research variable.	Graph correct; explanation identifies main impacts but lacks strong research question.	Graph present; explanation minimal or lacks detail on impacts.	Graph or explanation incomplete; unclear or inaccurate analysis.	Missing graph or incorrect; no analysis provided.

Layout	Fully meets all formatting requirements (font, margins, page limits, separate pages, graph size); report is polished and easy to follow.	Meets most formatting requirements; minor issues in layout or clarity.	Formatting mostly correct but with several issues; presentation adequate.	Formatting requirements not fully met; disorganized layout.	